

Part 2

The qualifying event for the 800m race is about to take place. The runners' times are recorded as they complete the first 400m lap and again for the second 400m lap. Runners who complete each lap in 58 seconds or less will qualify for the next stage of the championships. Today there will be a minimum of 2 and maximum of 8 runners in each race.

The program requires the following inputs:

- a valid number of runners
- the name of each runner
- the time taken to run the first lap
- the time taken to run the second lap

The output from the program should display each runner's name and indicate whether they have qualified for the next event.

An example is provided below

Runner Name	Qualified
D Haggerty	Yes

Your task is to create software to determine who will qualify for the next stage of the championships.

The top level algorithm is shown below. Steps 4 and 7 have been refined.

Pseudocode

MAIN STEPS

1. Get valid number of runners
2. Loop for number of runners
3. Get runner's name
4. Get a time for each lap
5. Decide if runner qualifies for next event
6. End loop
7. Display results

REFINEMENTS

4. Get a time for each lap
- 4.1 Get time for first lap
- 4.2 Get time for second lap

7. Display results
- 7.1 Display headings
- 7.3 Loop for number of runners
- 7.4 Display runner's name
- 7.5 Display qualifying decision
- 7.6 End loop

Tasks					Evidence required																				
1	Refine the following parts of the algorithm: • Get valid number of runners (step 1) • Decide if runner qualifies for next event (step 5) (NOTE: <i>all refinements must include an algorithm and not simply use a feature of an event-driven language.</i>)				Pseudocode for steps 1 and 5																				
2	Create a program that matches the refined algorithm.				Listing of program																				
3	Copy and complete the test table below. <table><tr><th>Runner Name</th><th>First 400m</th><th>Second 400m</th><th>Qualified</th></tr><tr><td>D Haggerty</td><td>56</td><td>58</td><td></td></tr><tr><td>S Ahmed</td><td>55</td><td>61</td><td></td></tr><tr><td>L Greenan</td><td>58</td><td>58</td><td></td></tr><tr><td>D Quinton</td><td>59</td><td>59</td><td></td></tr></table>				Runner Name	First 400m	Second 400m	Qualified	D Haggerty	56	58		S Ahmed	55	61		L Greenan	58	58		D Quinton	59	59		Set of test data
Runner Name	First 400m	Second 400m	Qualified																						
D Haggerty	56	58																							
S Ahmed	55	61																							
L Greenan	58	58																							
D Quinton	59	59																							
4	Test your program using the above test data.				Printed output																				
5	Test the number of runners using exceptional data.				Printed output																				